



Profile

Technology leader and software engineer/architect with 20+ years delivering complex systems across both the private and public sectors in domains such as climatology, quantum instrumentation, cryptography, blockchain, finance & telecoms. At home setting technical strategy, advising c-suite executives and shaping the processes that let engineering organisations scale. Track record of building a culture of trust where exceptional teams flourish. Member of the Institute of Electrical and Electronics Engineers (IEEE) and the Central Bank Research Association.

Experience

2025 – Present · concurrent

Executive Producer

Hitotoki Collective · Kyoto, Japan

Co-founder of an art collective that stages live multi-media performances in Kyoto temples. At the heart of each performance lies an improvised dialogue between a painter-calligrapher and a small ensemble of musicians; each piece may equally be read as an offering to the spirits of place. As producer, coordinating vision, funding, dissemination and wider cultural engagement.

2021 – Present

Senior Software Consultant

QZabre @ ETH · Zurich, Switzerland

Driving software at an ETH spin-off that builds nanoscale microscopes based on quantum magnetometry, supporting the team's growth from 3 to 8 engineers as shipments scaled to many dozens of devices per year. Guided the shift to a lightweight, structured agile process; designed and built controllers for photonic and magnetic devices; redesigned the desktop application; and championed the adoption of Rust across the in-house QS3 platform. Advising both the engineering team and the executive suite.

2025 – 2026 · concurrent

Head of Technical Strategy

Inversed Tech · Madrid, Spain

A hybrid role across an R&D company specialising in cryptography and distributed systems. As Rust engineer, built the genesis protocol and tooling for a biometric identity platform based on secure multi-party computation. As technical product owner, steered customer-facing projects including a regulated privacy-focused stablecoin and a privacy-preserving loyalty programme. As business development lead, represented the company at industry events, shaped strategy and tracked emerging trends.

2022 – 2024

Head of R&D

Casper Association · Zug, Switzerland

Led an 11-person R&D team in a complex, high-visibility blockchain environment. Partnered with the Head of Engineering to fold research proofs of concept into the core product, and worked with a team of macro-economists to design a blockchain-enabled core banking platform built around an emerging financial standard. Hands-on Rust throughout, applying zero-knowledge cryptography to deliver a ZK rollout sustaining 4,000 transactions per second and a ZK light client.

2009 – 2022 · concurrent

Software Consultant

Institut Pierre Simon Laplace @ La Sorbonne · Paris, France

Worked at the heart of climate research, extending the IPSL's HPC simulation environment with a real-time, event-driven messaging platform powering automated documentation, live monitoring, performance analytics and machine intelligence. Technical lead of es-doc.org, an international climate-model documentation project serving 35 institutes and over 1,000 users, and a driver of process change across the institute.

2019 – 2021

Software Consultant

Prove AI · Zug, Switzerland

Part of the global team that delivered a layer-1 blockchain platform built on DAG-based proof-of-stake consensus. Designed and implemented an event-driven fuzz-testing platform spanning 100 nodes that surfaced 23 consensus flaws, and streamlined platform integration for external partners. Also led the professional-services team and mentored executives.

2018 – 2019

Software Architect

Trinkler Software · Zug, Switzerland

Helped build a decentralised finance ecosystem for a Zug start-up. Delivered a GraphQL trading API, a feature-rich command-line application, a multi-chain digital wallet and a Vue.js front end, while mentoring junior colleagues and reviewing ecosystem design.

2008 – 2013

Software Architect

NTT Data · Fribourg, Switzerland

Drove technology and process change in the Swiss arm of a global IT services company: introduced agile practice — Scrum, XP, CI/CD — designed a .NET SOA framework and mentored a seven-person development team.

2007 – 2008

Software Architect

British Telecom · Dublin, Ireland

Designed and led development of an interoperable B2B order-processing web service, applying .NET 3.0, domain-driven design and SOA. Also designed a rules-based workflow engine.

2006 – 2007

Software Engineer

DeCare Systems · Cork, Ireland

Designed and led development of a high-volume, real-time enrolment service for a US insurance provider, using an n-tier architecture with WCF communication between tiers.

2006

Software Engineer

SnapOn Diagnostics · Cork, Ireland

Migrated C++ systems to .NET 2.0 under an agile process, and built web services sharing real-time automotive diagnostics across a suite of listening applications. Scrum, XP and TDD throughout.

2005

Software Engineer

Classic Car Collection · Paris, France

Full-stack .NET engineer on a sales and management information system, covering architecture and design, web development, testing and deployment.

1989 – 2001

Earlier Career

Securicor Systems · SSRADU · 3S Electronic Publishing · Customer Interface · Cap Gemini — UK

Software engineering and architecture roles across the UK, spanning n-tier client-server systems, COM frameworks in Visual Basic, database design and technical authorship. Highlights include architecting central London's traffic-violation adjudication system at the head of a twelve-person team (Securicor) and delivering Europe's largest IBM AS-400 project (Cap Gemini).

Skills

LANGUAGES & FRAMEWORKS

C#PythonRustTypeScriptBashReactVueSvelte

ARCHITECTURE

BlockchainEvent Driven ArchitectureMessage BrokersMicro-ServicesAgentic Systems

AGENTIC (CCA-F CURRICULUM)

Agentic ArchitectureClaude Agent SDKModel Context Protocol (MCP)Prompt EngineeringStructured OutputTool Design & IntegrationContext & Reliability

DATA & MESSAGING

PostgreSQLMongoDBElasticsearchOracleSQL ServerRabbitMQ

DEVOPS & PLATFORMS

DockerKubernetesTerraformAnsibleAWS CloudLinux

PROCESS

AgileScrumXPKanbanTDDCI/CDGitHub

Education

2026

Claude Certified Architect Foundation (in progress)

Anthropic

2008

Microsoft Certified Software Architect

Microsoft Campus, San Francisco, USA

1989

B.Sc. Economics & Operational Research

University of Leeds, UK

Other

An Englishman by birth, granted leave to dwell among the Swiss mountains of Zug; married, and entrusted with two children — a duty greater than any commission. The languages came not from academies but from years of labour among other peoples: English and French worn smooth by daily use, Italian learned willingly,

German enough for honest work. The road has led through Japan and China, through India and the Middle East, across North and Central America — and everywhere the same lesson: a man is measured not by what he owns but by what he withstands. What remains after the day's work is simple and sufficient: the contest of tennis, cricket and rugby, the discipline of swimming and the bicycle; song, the writing of words, the long memory of history, and art — which alone does not lie.